HUFFMAN'S ALGORITHM

- **Draw the optimal Huffman Tree for the given string below.** 22104487 이경원

"Why not change the world?"

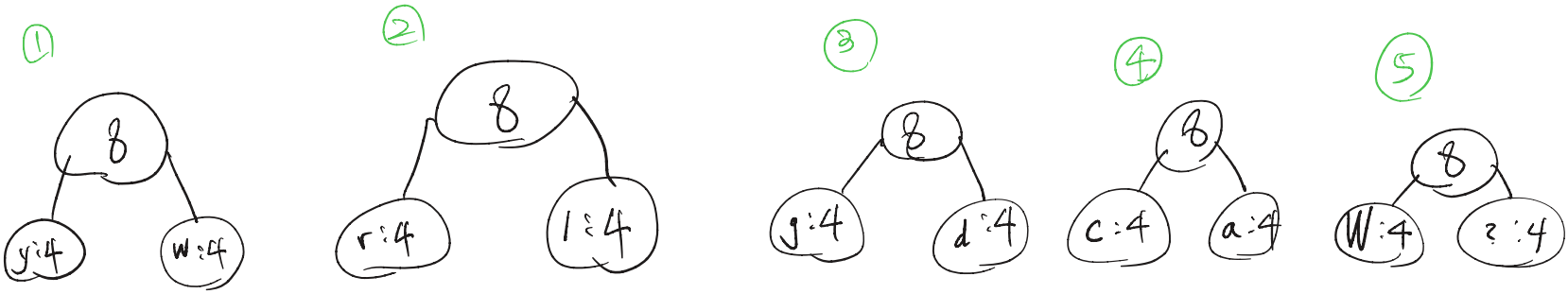
(Note: Treat uppercase/lowercase letters, spaces, and the question mark as distinct characters.)

- **Generate the corresponding prefix code table based on your tree.**
- **Encode the entire phrase into a binary bitstream using the table.**

- **Requirements**
  - ✓ You must show the entire process of deriving your results.
  - ✓ Assign binary codes (0 for left, 1 for right) to each character based on the tree structure.
  - ✓ Due: 4/14(Tue) 2:30pm

	(Space)	?	W	a	c	d	e	g	h	l	n	o	r	t	w	y
Count	4	1	1	1	1	1	2	1	3	1	2	2	1	2	1	1
Frequency	16	4	4	4	4	4	8	4	12	4	8	8	4	8	4	4

make tree 1.

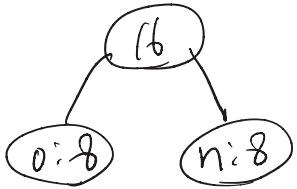


make tree 2

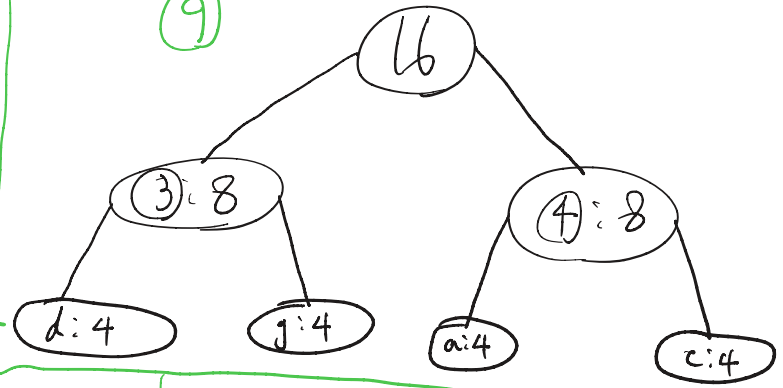
	(Space)	⑤ (?, W)	④ (a, c)	③ (d, g)	e	h	② (l, r)	n	o	t	① (w, y)
Frequency	16	8	8	8	8	12	8	8	8	8	8

⑨ ⑧ ⑦ ⑥ ⑩

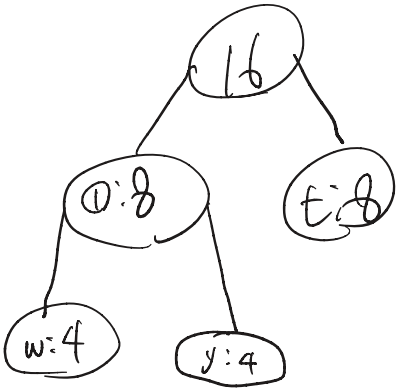
⑦



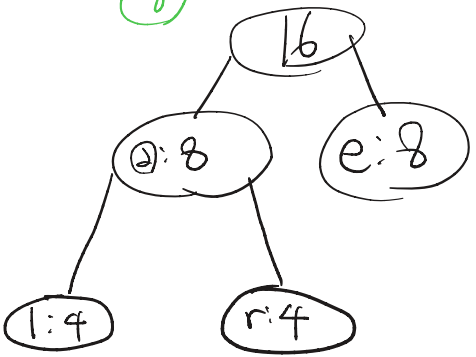
⑨



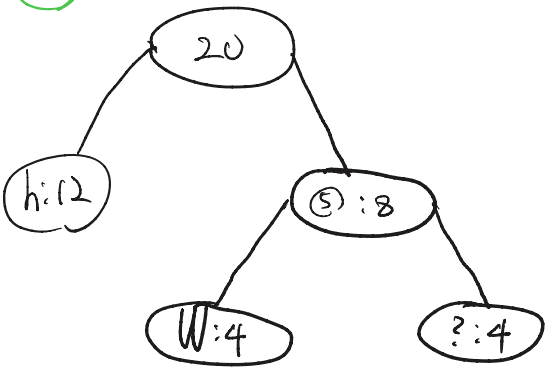
⑥



⑧

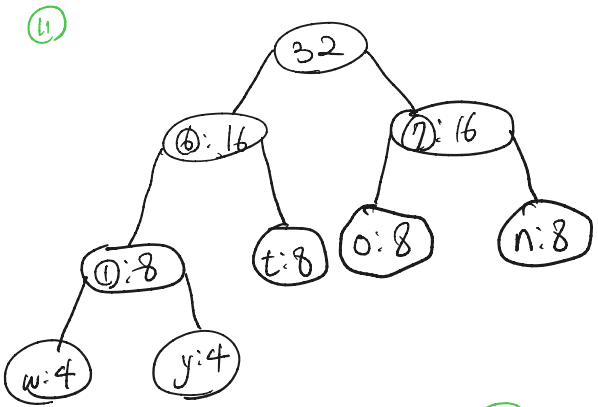


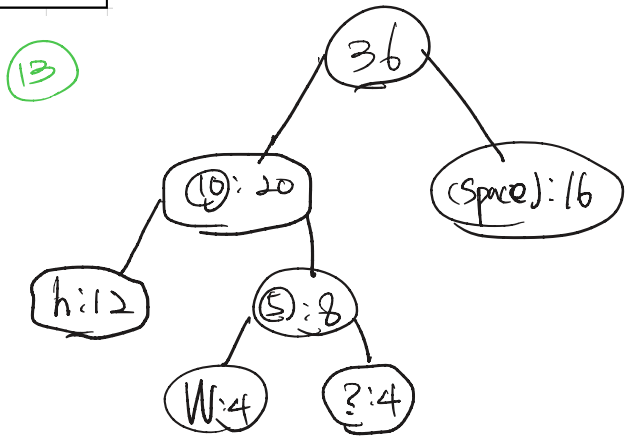
⑩

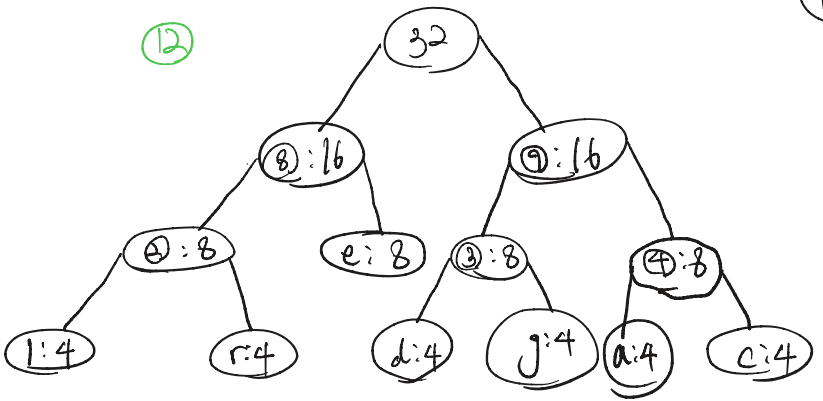


make-tree 3

	(Space)	⑩ (h, ⑤)	⑨ (③, ④)	⑧ (②, e)	⑦ (o, n)	⑥ (①, t)
Frequency	16	20	16	16	16	16

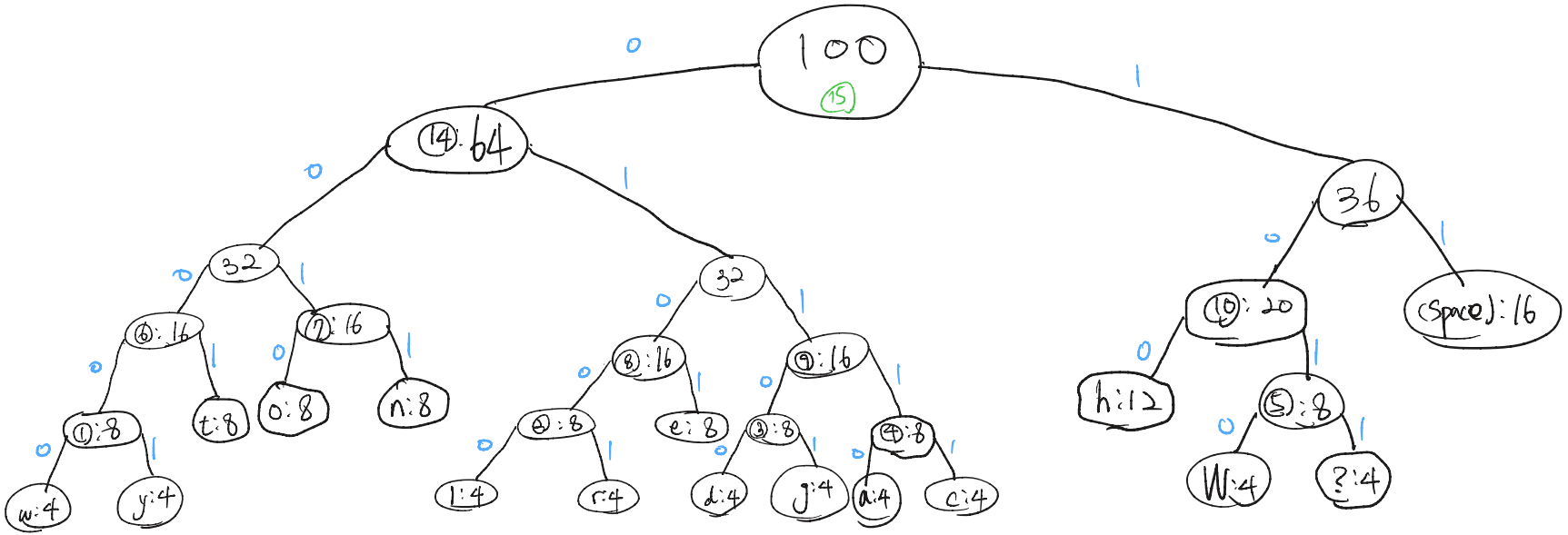






make tree 4

	⑬ (⑩, (Space))	⑫ (⑧, ⑨)	⑪ (⑥, ⑦)
Frequency	36	32	32



Prefix Code Table

(Space)	?	W	a	c	d	e	g
11	1011	1010	01110	01111	01100	0101	01101
h	l	n	o	r	t	w	y
100	01000	0011	0010	01001	0001	00000	00001

Why : 1010100 00001
(space) : 11
not : 001001010 0001
(space) : 11
change : 01111100011000110101010 1
(space) : 11
the : 0001100010 1
(space) : 11
world : 00000001001001010001100
? : 1011

Encode the entire phrase into a binary bitstream.

1010 100 00001 11 0011001 0000 1 11 01111 100 01110 0011 01101010 1 11 0001 100010 1 11 00000001001001010001100 1011

W h y v n o t v c h a n g e v t h e v w o r l d ?